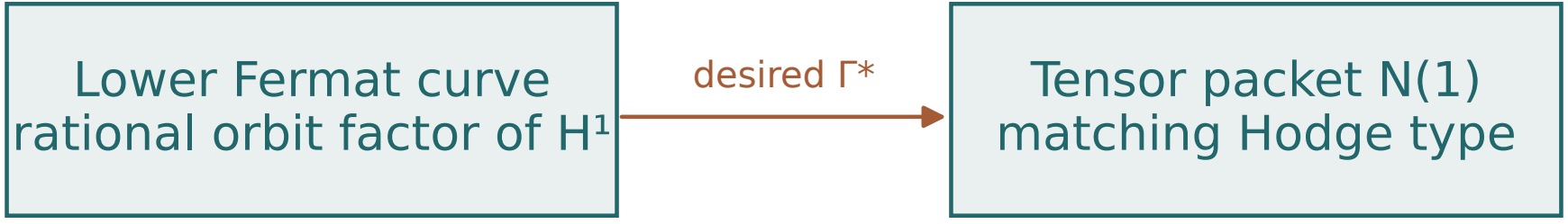

Chow correspondence still required

Character covariance + a nonzero projected coefficient
+ rational reconstruction of the whole orbit

CM sign matching identifies a candidate. It does not construct Γ.